

Total Deep Variation Regularization for Improved Iterative Quantitative Susceptibility Mapping (TDV-QSM)

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Synopsis

Quantitative Susceptibility Mapping (QSM) is an ill-posed inverse problem. Traditionally, it is solved by minimization of a functional. Regularization terms may be interpreted as a denoising process. Many state-of-the-art methods are based on Total Variation regularization terms, with great success. With the advent of Deep Learning, new regularization strategies have been derived from training datasets. Total Deep Variation (TDV) is a recently proposed technique, showing impressive results. We applied a pre-trained TDV network as a denoising step in an iterative QSM solver. Results show improved error metrics for synthetic brain phantoms and enhanced in-vivo reconstructions, compared to Total-Variation-based algorithms.

PURPOSE

Inferring the underlying tissue magnetic susceptibilities from the gradient-echo phase is an ill-posed inverse problem¹, known as Quantitative Susceptibility Mapping (QSM). QSM is typically solved by minimization of a cost functional with a data fidelity term and one or more regularization terms². Whereas the data fidelity term enforces consistency of the solutions with the acquired data and models the noise distribution, regularization terms enforce a-priori knowledge regarding the solution, such as smoothness³ or sparsity. Total Variation (TV)^{4,5} and terms derived from it (by introducing local weights⁴) or extensions such as Total Generalized Variation, TGV^{5,6} are popular QSM regularizers, achieving the highest scores in the 2019 QSM Reconstruction Challenge (RC2)⁷. Recent advances in Deep Learning⁸ have provided a new approach to build regularization terms. In particular, variational networks have been successfully applied to denoising and many other linear inverse problems^{9,10}. Total Deep Variation (TDV)¹¹ is a state-of-the-art variational network, based on the 'Fields of Experts' model¹² (a generic framework to learn image priors), but requiring only a fraction of its parameters. This makes TDV easier to train and more generalizable to different problems¹¹. Here, we used a pre-trained (BSDS400 dataset¹³) TDV network (<https://github.com/VLOGroup/tdv>), designed to remove Gaussian noise from 2D color images, as a proximal (regularization) step within each iteration for QSM reconstruction and evaluated whether it reduced noise and streaking artifacts in the resulting susceptibility maps.

METHODS

In general, we may describe the functional of the QSM inverse problem as $\argmin_{\chi} F(\chi, \phi) + \alpha R(\chi)$, where $F(\chi, \phi)$ is the data fidelity term between the susceptibility distribution χ and the measured phase ϕ (such as the linear³ or nonlinear L2-norm^{4,5}). $R(\chi)$ is the regularization term, with α the Lagrangian weight that balances both terms. To use TDV in an efficient⁵ proximal step, we introduced the $z = \chi$ variable to split the regularization term from the data fidelity term using the Alternating Directions of Multipliers Method (ADMM) framework^{5,14}. The augmented functional becomes:

$$\argmin_{\chi, z} F(D\chi, \phi) + \alpha R(z) + \frac{\mu}{2} \|\chi - z + s\|_2^2$$

with μ an additional Lagrangian weight, and s a Lagrange multiplier. Each subproblem is solved consecutively, until convergence. First we solve the χ subproblem as previously described in the FANSI⁵ algorithm. Now, the z subproblem becomes a TDV denoising problem, with input $\chi_{k+1} + s_k$ at each iteration k , allowing us to use the pre-trained denoising network. The regularization weight α becomes the scale parameter in TDV that adapts the network to the input noise level. μ balances the regularization and the data fidelity terms. Finally, we update $s_{k+1} = s_k + \chi_{k+1} - z_{k+1}$ and all other Lagrange multipliers.

The pre-trained TDV network for 2D color images was extended to 3D images by inputting groups of three consecutive slices as red, green and blue channels of 2D color images, until the whole volume was denoised. We compared reconstructions obtained by TDV regularization with FANSI-TV⁵ (which achieved the best RMSE in RC2, Stage 2)⁷ and FANSI-TGV⁵. We used two synthetic brain models to compare reconstructions: 1) A forward simulation (SNR=100) from the COSMOS-based 2016 QSM Challenge dataset (RC1)¹⁵, and 2) the SIM2SNR1¹⁶ field map from RC2. We also compared the reconstructions in a healthy female volunteer scanned for another study¹⁷ at 3T (Achieva, Philips Healthcare, NL) using a 32-channel head coil and a 3D GRE sequence with 5 echoes, TE1/ΔTE/TR=3/5.4/29ms, flip angle=20°, matrix size = 240×240×144, and 1 mm3 isotropic voxels. Multi-echo combination was performed by nonlinear fitting⁴. Unwrapping was performed using SEGUE¹⁸. Background field removal was performed by the Laplacian Boundary Value (LBV)¹⁹ method and then by Weak Harmonics QSM (WH-QSM)²⁰ to remove any residual background fields. We used bespoke masking based on R2* maps to remove inflow effects in the reconstructions. The root mean squared error (RMSE), QSM-tuned structural similarity index²¹ (XSIM), and RC2-specific metrics⁷ were used to compare the phantom susceptibility maps.

RESULTS

Optimal RMSE reconstructions and error maps using RC1 are shown in Figure 1. TDV shows better depiction of cortical areas (a-d) and less staircasing (a) and streaking artifacts (error maps) than TV and TGV. TDV converged with slightly fewer iterations than TV and TGV (Figure 2), achieving better RMSE and XSIM scores (Figure 1). TDV results in RC2 show better depiction of the veins (Figure 3, b-d) and streaking artifact suppression (a) relative to TV and TGV. TDV converged with fewer iterations and was more stable (Figure 4), performing better in almost all error metrics. In-vivo reconstructions are shown in Figure 5. TDV shows increased contrast in some deep structures and between white matter and grey matter (a-c), and some attenuation of noise and streaking near the vessels (d-h).

DISCUSSION

In all our experiments, optimal TV and TGV reconstructions are almost identical. TDV results show promising, albeit subtle, improvements over TV and TGV, despite the increased computation time. Our implementation is not optimal, and may be improved in terms of performance and computation time by retraining the TDV network with 3D data. Retraining may also target suppression of QSM streaking artifacts.

CONCLUSION

We have demonstrated that, even using a pre-trained network, Total Deep Variation denoising may be used for QSM and it achieved improved reconstructions compared to state-of-the-art TV and TGV regularized algorithms. Future TDV developments will focus on QSM-specific retraining and solver optimization.

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Figures

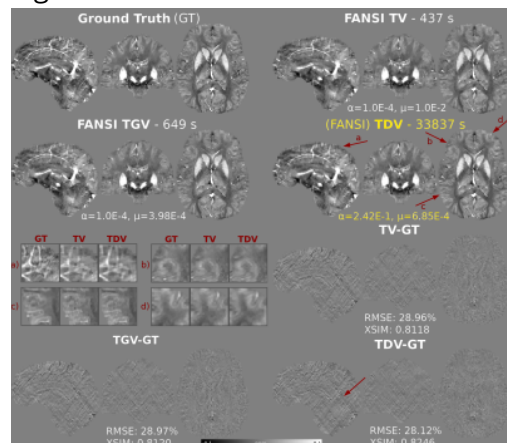


Figure 1: Optimal reconstructions and error maps of COSMOS-based forward simulations (RC1) for all regularization methods (TDV, TV and TGV). A sagittal, coronal and axial slice are shown. All methods were terminated after 500 iterations. TDV shows better depiction of cortical areas and less staircasing and streaking artifacts (highlighted with red arrows) than TV and TGV and achieves better RMSE and XSIM scores. Detailed comparisons between TV and TDV are shown for each labeled region (a-d).

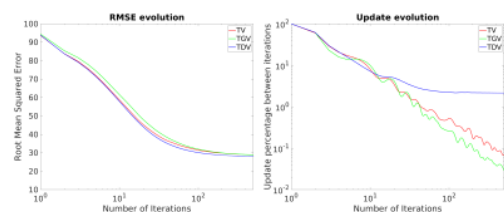


Figure 2: RMSE and update convergence for all regularization methods (TDV, TV and TGV) on the RC1 numerical phantom. The update convergence is defined as: $100||\chi_k - \chi_{k-1}||/||\chi_k||$ where k is the iteration number. TDV converges very slightly faster than TV and TGV.

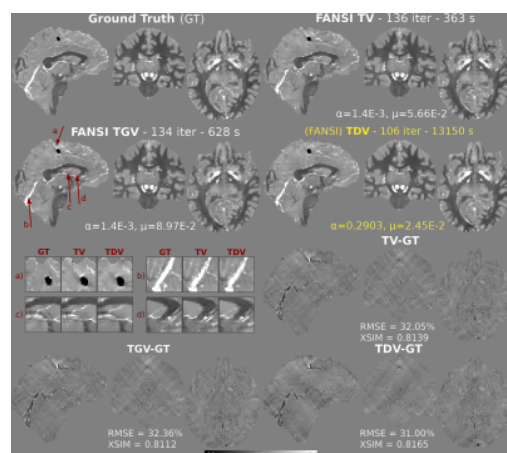
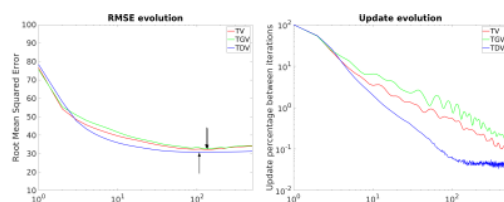


Figure 3: Optimal reconstructions of the RC2 numerical phantom for all regularization methods (TDV, TV and TGV). TDV results show better depiction of the veins and streaking artifact suppression relative to TV and TGV (relevant areas highlighted with red arrows).



Number of iterations										Number of iterations									
RMSE	RMSE	RMSE	RMSE	RMSE	RMSE	RMSE	RMSE	RMSE	RMSE	RMSE	RMSE	RMSE	RMSE	RMSE	RMSE	RMSE	RMSE	RMSE	RMSE
TV	32.05	32.05	32.05	32.05	32.05	32.05	32.05	32.05	32.05	32.05	32.05	32.05	32.05	32.05	32.05	32.05	32.05	32.05	32.05
TGV	32.36	32.36	32.36	32.36	32.36	32.36	32.36	32.36	32.36	32.36	32.36	32.36	32.36	32.36	32.36	32.36	32.36	32.36	32.36
TDV	31.06	31.06	31.06	31.06	31.06	31.06	31.06	31.06	31.06	31.06	31.06	31.06	31.06	31.06	31.06	31.06	31.06	31.06	31.06

Figure 4: RMSE and update convergence for all regularization methods (TDV, TV and TGV) on the RC2 numerical phantom (Top). The update convergence is defined as: $100||\chi_k - \chi_{k-1}||/||\chi_k||$ where k is the iteration number. An extended set of RC2 error metrics for all methods is shown below. TDV converged faster and was more stable, performing better in almost all error metrics.

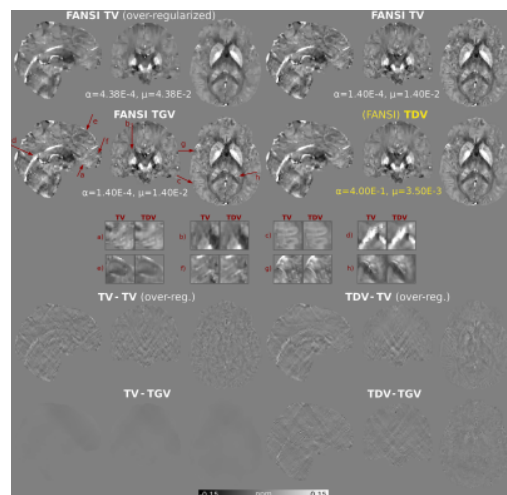


Figure 5: Optimal reconstructions (L-curve analysis) for TDV, TV and TGV in vivo. An over-regularized TV solution is also shown to facilitate assessment of streaking artifacts and the sharpness of fine details. Relative to TV and TGV, TDV shows larger differences for deep brain structures (due to increased contrast), and smaller differences for cortical areas (indicating stronger streaking artifact suppression). Areas highlighted with arrows (a-h) show that TDV clearly depicts vessels and other structures, while noise is suppressed without noticeable artifacts.